NexTask Platform — Gateway API Integration Guide

Base URL: https://nextaskgw.automationedge.com | **Auth Protocol:** Token & API Key Authentication

Authentication Architecture & Workflow Overview:
• **Step 1 (Authenticate):** Send administrative credentials to POST /api/authorize.
• **Step 2 (Obtain Key):** The API returns your active api_key and session token in the response body. *(Keys are generated and returned in the response rather than hardcoded).*
• **Step 3 (Invoke APIs):** Pass the received key as X-API-Key: <api_key> in HTTP headers for all subsequent API calls.
• **Admin Credentials:** Username: **admin** | Password: **admin123**

1. Authorization API (/api/authorize)

Method: POST / GET **Endpoint:** https://nextaskgw.automationedge.com/api/authorize

Authenticates administrative credentials and dynamically returns the active api_key, x_api_key, and token. Call this endpoint first to obtain the API key required for all subsequent requests.

Required Request Headers:

Header Name	Sample / Format	Requirement	Notes & Purpose
Content-Type	application/json	Required	Specifies JSON payload format
Authorization	Basic YWRtaW46YWRtaW4xMjM=	Optional	HTTP Basic Auth header alternative (base64 admin:admin123)

Request Body (JSON):

```
{
  "username": "admin",
  "password": "admin123"
}
```

Success Response (200 OK — Returns Dynamic API Key):

```
{
  "status": "success",
  "authorized": true,
  "http_code": 200,
  "message": "User authenticated successfully. Use the provided api_key in the headers of other APIs.",
  "api_key": "<dynamic_api_key_returned_here>",
  "x_api_key": "<dynamic_api_key_returned_here>",
  "token": "4a7f9c2d1b8e...",
  "token_type": "Bearer",
  "expires_in": 86400,
  "user": {
    "username": "admin",
    "role": "admin"
  }
}
```

cURL Command (Separate Execution):

```
curl -X POST https://nextaskgw.automationedge.com/api/authorize \
  -H "Content-Type: application/json" \
  -d '{"username": "admin", "password": "admin123"}'
```

2. Push Records API (POST /api/alerts)

Method: POST (or PUT) **Endpoint:** https://nextaskgw.automationedge.com/api/alerts

Ingests ticket/task records directly into the NexTask database. Records immediately appear in the operations database and platform views. **All 17 fields below are strictly required** as per the data ingestion specification.

Required Request Headers:

Header Name	Header Value	Requirement	Description
Content-Type	application/json	Required	Must be application/json
X-API-Key	<api_key_from_authorize_response>	Required	API key received from /api/authorize response
Authorization	Bearer <token_from_authorize_response>	Alternative	Bearer token alternative to X-API-Key

Required Data Schema (All 17 Fields Mandatory):

#	Field Name	Req.	Sample Value	Description
1	**task_id**	Yes	TASK1001	Unique ticket / task identifier
2	**issue_description**	Yes	Network connectivity failure	Detailed fault description
3	**issue_type**	Yes	Network Failure	Category / Alarm classification
4	**severity**	Yes	Critical	Critical / Major / Minor / Normal
5	**task_status**	Yes	Dispatch	Dispatch / Assigned / In Progress
6	**alarm_status**	Yes	Active	Active / Cleared
7	**site_number**	Yes	SITE001	Branch / ATM Site ID
8	**created_date**	Yes	2026-11-03T10:30:00+03:00	ISO 8601 creation timestamp
9	**technician_name**	Yes	Ali	Assigned field engineer name
10	**technician_mobile**	Yes	+966500000001	Technician phone number (+E.164)
11	**technician_preferred_lang**	Yes	English	English / Arabic
12	**teamleader_name**	Yes	Mohammed	Team Leader supervisor name
13	**teamleader_mobile**	Yes	+966500000002	Team Leader phone number
14	**teamleader_preferred_lang**	Yes	Arabic	Arabic / English
15	**manager_name**	Yes	Khalid	Escalation Manager name
16	**manager_mobile**	Yes	+966500000003	Escalation Manager phone number
17	**manager_preferred_lang**	Yes	English	English / Arabic

2. Push Records API (Continued)

Sample Request Body (JSON):

```
{
  "tickets": [
    {
      "task_id": "TASK1001",
      "issue_description": "Network connectivity failure",
      "issue_type": "Network Failure",
      "severity": "Critical",
      "task_status": "Dispatch",
      "alarm_status": "Active",
      "site_number": "SITE001",
      "created_date": "2026-11-03T10:30:00+03:00",
      "technician_name": "Ali",
      "technician_mobile": "+966500000001",
      "technician_preferred_lang": "English",
      "teamleader_name": "Mohammed",
      "teamleader_mobile": "+966500000002",
      "teamleader_preferred_lang": "Arabic",
      "manager_name": "Khalid",
      "manager_mobile": "+966500000003",
      "manager_preferred_lang": "English"
    }
  ]
}
```

Success Response (200 OK):

```
{
  "status": "success",
  "message": "Processed 1 record(s): 1 stored successfully, 0 failed",
  "records_count": 1,
  "success_count": 1,
  "failed_count": 0,
  "tickets": [
    {
      "index": 0,
      "task_id": "TASK1001",
      "ticket_id": "TASK1001",
      "status": "stored",
      "site_number": "SITE001",
      "severity": "Critical",
      "technician": "Ali (+966500000001)"
    }
  ]
}
```

cURL Command (Separate Execution):

```
curl -X POST https://nextaskgw.automationedge.com/api/alerts \
  -H "Content-Type: application/json" \
  -H "X-API-Key: <api_key_from_authorize_response>" \
  -d @payload.json
```

3. Update Records API (PUT /api/alerts/update)

Method: PUT (or POST) **Endpoint:** https://nextaskgw.automationedge.com/api/alerts/update

Updates existing tickets in the database. Supports technician reassignments, movement updates (e.g. Departed, On Site), and ticket closures / alarm clearances (Closed, Cleared). Accepts single objects or batch arrays under the updates key.

Required Request Headers:

Header Name	Header Value	Requirement	Description
Content-Type	application/json	Required	Must be application/json
X-API-Key	<api_key_from_authorize_response>	Required	API key received from /api/authorize response
Authorization	Bearer <token_from_authorize_response>	Alternative	Bearer token alternative to X-API-Key

Field Specification & Business Rules:

Field Name	Requirement	Notes & Validation Rules
task_id	Required	The ticket ID to update (e.g. TASK1001)
update_type	Required	Must be either 'REASSIGNMENT' or 'STATUS_UPDATE'
technician_name	Required for Reassignment	New field technician name
technician_mobile	Required for Reassignment	New field technician phone number (+E.164)
technician_preferred_lang	Required for Reassignment	Preferred communication language (English / Arabic)
task_status	Optional	e.g. 'Departed', 'On Site', 'Resolved', 'Closed'
alarm_status	Optional	e.g. 'Active', 'Cleared'
severity	Optional	Updated severity classification if changed

Sample Request Body (JSON):

```
{
  "updates": [
    {
      "task_id": "TASK1001",
      "update_type": "REASSIGNMENT",
      "technician_name": "Faisal",
      "technician_mobile": "+966500000007",
      "technician_preferred_lang": "Arabic"
    },
    {
      "task_id": "TASK1002",
      "update_type": "STATUS_UPDATE",
      "task_status": "Departed"
    },
    {
      "task_id": "TASK1003",
      "update_type": "STATUS_UPDATE",
      "task_status": "Closed",
      "alarm_status": "Cleared"
    }
  ]
}
```

cURL Command (Separate Execution):

```
curl -X PUT https://nextaskgw.automationedge.com/api/alerts/update \
  -H "Content-Type: application/json" \
  -H "X-API-Key: <api_key_from_authorize_response>" \
  -d @updates.json
```

4. Voice Calling & Escalation API (/api/call)

Method: POST /api/call & GET /api/call/{ticket_id} **Endpoint:** https://nextaskgw.automationedge.com/api/call

Triggers automated Twilio outbound voice calls to field technicians with multi-lingual IVR and keypad DTMF acknowledgment. If the technician does not answer after 2 call attempts, the platform automatically executes escalation:

Automated Escalation Architecture (When Technician Misses 2 Calls):

- **1. Retry Mechanism:** Call #1 is placed. If unanswered/busy, system waits retry interval (e.g. 60s) and places Call #2.
- **2. Instant SMS Alerts:** If Call #2 is unanswered, automated SMS notifications are immediately dispatched via Twilio to both the Team Leader and Manager.
- **3. Supervisory Voice Call:** An automated IVR call is placed to the Team Leader asking: *'Do you want to reassign this ticket? Press 1 for YES, Press 9 for NO.'*
- **4. Manager Fallback:** If the Team Leader does not answer, the call cascades to the Escalation Manager.

Sample Call Trigger Request Body (POST /api/call):

```
{
  "alert_id": "ALM-1001",
  "ticket_id": "TASK1001",
  "phone_number": "+966500000001",
  "site_details": "Olaya Branch ATM, Riyadh",
  "technician_name": "Ali",
  "technician_language": "en",
  "team_leader_number": "+966500000002",
  "team_leader_name": "Mohammed",
  "manager_number": "+966500000003",
  "manager_name": "Khalid",
  "severity": "Critical",
  "description": "Network connectivity failure",
  "ack_digit": "1",
  "reject_digit": "9"
}
```

cURL Command (POST /api/call):

```
curl -X POST https://nextaskgw.automationedge.com/api/call \
  -H "Content-Type: application/json" \
  -H "X-API-Key: <api_key_from_authorize_response>" \
  -d @call_payload.json
```

4. Voice Calling & Escalation API (Continued)

Response Schema for GET /api/call/{ticket_id}: Returns full real-time audit details including DTMF key pressed, technician call status, Twilio voice calls array, and Twilio SMS notification dispatch logs (**sms_messages**).

Complete Real-Time Response Payload (GET /api/call/{ticket_id}):


```
{
  "ticket_id": "TASK1001",
  "alert_id": "ALM-1001",
  "site_name": "Olaya Branch ATM, Riyadh",
  "severity": "Critical",
  "status": "Assigned",
  "acknowledged": true,
  "dtmf_pressed": "1",
  "ack_number_pressed": "1",
  "technician_call_status": "answered",
  "visit_status": "accepted",
  "final": true,
  "call_workflow": {
    "state": "ACKNOWLEDGED",
    "final": true,
    "acknowledged": true,
    "acknowledged_via": "voice",
    "acknowledged_at": "2026-11-04T09:01:22+03:00",
    "technician_response": "acknowledged",
    "technician_picked_up": true,
    "escalation": null,
    "response_summary": "Technician Ali acknowledged via voice call (DTMF 1)"
  },
  "calls": [
    {
      "id": 1,
      "role": "technician",
      "workflow": "uc1_technician",
      "attempt_no": 1,
      "recipient_name": "Ali",
      "to_number": "+966500000001",
      "language": "en",
      "call_sid": "CA1234567890abcdef...",
      "status": "completed",
      "outcome": "acknowledged",
      "dtmf": "1",
      "duration_sec": 34,
      "picked_up": true
    }
  ],
  "sms_messages": [
    {
      "id": 1,
      "workflow_id": "WF-TASK1001-A1",
      "ticket_id": "TASK1001",
      "recipient_role": "team_leader",
      "recipient_name": "Mohammed",
      "mobile_number": "+966500000002",
      "message_type": "Escalation",
      "provider_message_id": "SM9a8b7c6d5e4f3a2b1c0d9e8f7a6b5c4d",
      "template_name": "sms_alert_notification",
      "delivery_status": "sent",
      "sent_at": "2026-11-04T09:02:16+03:00",
      "failure_reason": null
    }
  ],
  "status_url": "/api/call/TASK1001"
}
```


cURL Command (Poll Status):

```
curl -X GET https://nextaskgw.automationedge.com/api/call/TASK1001 \
  -H "X-API-Key: <api_key_from_authorize_response>"
```

5. HTTP Status Codes & Error Handling

All NexTask Gateway endpoints return standard HTTP status codes along with descriptive JSON error structures.

HTTP Code	Status Key	Common Scenario	Remediation / Corrective Action
200 OK	success	Request processed and stored	Operation completed successfully.
202 Accepted	accepted	Call scheduled / queued	Outbound telephony job initiated; poll status_url.
207 Multi-Status	partial	Some records succeeded, some failed	Inspect the 'tickets' or 'updates' array for per-record error items.
400 Bad Request	error	Missing required fields (e.g. 17 fields)	Check the error response message; supply all missing required fields.
401 Unauthorized	error	Missing or invalid X-API-Key header	Call /api/authorize with admin / admin123 and pass returned key in X-API-Key.
404 Not Found	error	Target task_id not found on update	Verify ticket ID was ingested via /api/alerts prior to updating.
409 Conflict	error	Duplicate active ticket in progress	Wait for active ticket workflow to complete or use /api/alerts/update.

API Endpoints Quick Reference

Endpoint	Method	Required Headers	Function / Description
/api/authorize	**POST**	Content-Type: application/json *(or Basic Auth header)*	Authenticates admin credentials; returns dynamic api_key and token.
/api/alerts	**POST**	Content-Type: application/json **X-API-Key:** <api_key>	Pushes new ticket records (17 mandatory fields) into the operations database.
/api/alerts/update	**PUT / POST**	Content-Type: application/json **X-API-Key:** <api_key>	Updates existing tickets (Technician Reassignment or Status / Closure Updates).
/api/call	**POST**	Content-Type: application/json **X-API-Key:** <api_key>	Triggers automated Twilio voice calls with DTMF acknowledgment and supervisory escalation.
/api/call/{ticket_id}	**GET**	**X-API-Key:** <api_key>	Polls real-time call outcomes, DTMF keys pressed, and sms_messages dispatch logs.

Integration Header Checklist for API Consumers:

1. **Always include** Content-Type: application/json in all POST and PUT requests.
2. **Obtain API Key:** Invoke POST /api/authorize once at application startup or session initialization.
3. **Attach Key:** Copy the returned api_key string and pass it in the header X-API-Key: <api_key> on every call to /api/alerts, /api/alerts/update, and /api/call.
4. **Alternative Header:** The gateway also accepts Authorization: Bearer <token> using the token field from the authorization response.